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RESEARCH ARTICLE

Human Daily Indoor Action (HDIA) Dataset: Privacy-Preserving Human Action Recognition Using Infrared Camera and Wearable Armband Sensors

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ABSTRACT Human Activity Recognition (HAR) plays a vital role in applications such as healthcare, smart homes, and robotics, particularly in supporting the elderly. However, most existing HAR datasets focus on general human activities and are typically collected using RGB cameras in controlled environments with fixed angles—conditions that limit their real-world applicability. In this study, we introduce the Human Daily Indoor Actions (HDIA) dataset, specifically designed to capture natural indoor activities performed by elderly individuals. The dataset includes 48 daily actions, recorded using non-invasive infrared (IR) cameras and wearable armband sensors positioned at various angles to ensure diverse and realistic activity representation. The use of IR sensors enhances privacy, making the dataset ethically suitable for long-term monitoring. To demonstrate its utility, we implemented a fusion-based HAR model that integrates data from both IR and Inertial Measurement Unit (IMU) sensors. This model achieves strong activity recognition performance while minimizing the risk of identity exposure. By focusing on privacy-aware data collection and the daily routines of elderly individuals, the HDIA dataset offers a valuable resource for advancing real-world HAR research.

INDEX TERMS Human action recognition (HAR), multimodal human action dataset (MHAD), IR, inertial sensors, electromyography signal, privacy protection.

I. INTRODUCTION

The objective of Human Activity Recognition (HAR) is to classify human activities based on the data captured by sensors. HAR can be applied across a wide range of domains, including surveillance [1], autonomous navigation systems [2], human-computer interaction [3], and

medicine. In these applications, there is an increasing need to recognize a wider range of everyday activities, such as sitting, standing, running, cycling, and answering the phone. To improve the accuracy of HAR, researchers have explored multimodal approaches that utilize different types of sensor data, including depth, infrared (IR), accelerometer, inertial measurement units (IMU), and gyroscopes. These sensors provide redundant yet complementary information, enabling a more comprehensive understanding of human activities.

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HAR methods can be broadly categorized based on the types of sensors employed: visual and non-visual approaches. Visual methods rely on video data captured by camera sensors to identify human activities. These approaches commonly utilize various three-dimensional (3D) video backbone networks, such as Pseudo 3D (P3D) [4], Two-stream Inflated 3D (I3D) [5], Separable 3D (S3D) [6], [7], (2+1)D [8], and the Temporal Shift Module (TSM) [9], to model and represent spatiotemporal patterns of activity. Widely used RGB video datasets for HAR include University of Central Florida 101 action classes (UCF101) [10], Human Motion Database with 51 classes (HMDB51) [11], Kinetics-400 [12], and Charades [13]. These datasets focus primarily on recognizing common activities from standard 2D video data, which contain rich visual details but lack depth information. To address this limitation, datasets such as Microsoft Research (MSR) Action 3D [14] and MSR DailyActivity 3D [15] provide both RGB and depth data, offering a more comprehensive understanding of human movements. Additionally, datasets like Infrared Action Recognition (InFAR) [16] and Indian Institute of Technology Roorkee Infrared Action Recognition (IITR-IAR) [17] incorporate thermal imagery captured by IR cameras to facilitate activity recognition under low-light or night-time conditions.

Non-visual HAR methods utilize a variety of sensors other than video cameras, such as accelerometers and gyroscopes commonly embedded in smartphones, smartwatches, and other wearable devices. To analyze the time-series signals captured by these sensors, sequence modeling architectures like Long Short-Term Memory (LSTM) networks [18] and Transformer [19] are frequently employed [20], [21]. Widely used datasets for non-visual HAR include Physical Activity Monitoring for Ageing People-2 (PAMAP2) [22], USC-HAD [23], and OPPORTUNITY [24]. Compared to visual methods, non-visual approaches offer advantages such as improved privacy protection and greater robustness to changes in lighting and environmental conditions.

The methods proposed in [33] and [34] leverage multimodal information from both visual and non-visual sensors to enhance the robustness and accuracy of HAR systems. These studies introduce sensor fusion architectures that aggregate features independently extracted from each modality, enabling more comprehensive activity representations. Several publicly available datasets support multimodal HAR by providing both visual and non-visual data, including University of Texas at Dallas Multimodal Human Action Dataset (UTD-MHAD) [29], Berkeley MHAD [30], Changzhou University-MHAD (CZU-MHAD) [31], and Continuous-MHAD (C-MHAD) [32]. These datasets have facilitated the development and evaluation of effective multi-modal fusion models. A summary of their key characteristics is presented in Table 1.

Although RGB camera images are commonly used for developing HAR systems, they raise significant concerns regarding the potential compromise of personal information. As RGB cameras capture human identities, there is an

inherent risk of violating privacy and identity protection. Moreover, most existing datasets were collected in controlled laboratory environments, which may not accurately represent the natural behavior of participants in real-world settings. To overcome these limitations, this study presents a novel multimodal indoor HAR dataset called the *Human Daily Indoor Actions* (HDIA) dataset.

Our HDIA dataset ensures a high level of personal information protection by utilizing IR cameras for visual data and wearable armband sensors for non-visual data. The dataset includes data on the daily activities of 47 participants living in a three-bedroom rental house. A total of 47 subjects participated in the experiments, with 48 distinct action categories labeled by human annotators. The dataset was collected in a minimally controlled environment, allowing participants to move freely throughout the house. This approach results in the HDIA dataset reflecting natural human activities observed in everyday life.

Table 1 highlights the distinctive features of our HDIA dataset in comparison to several publicly available HAR datasets. Notably, the HDIA dataset includes the largest number of subjects and activity classes among existing multimodal HAR datasets. In addition, it uniquely provides Electromyography (EMG) signals.

To facilitate further research, we have established a public benchmark based on the HDIA dataset. Alongside this, we provide a baseline HAR method that employs Transformer attention to fuse features extracted from two modalities. This benchmark enables comparative evaluation of fusion-based HAR techniques and serves as a foundation for future research in multimodal activity recognition.

A summary of the contributions is as follows:

- We released a novel dataset for HAR that is particularly well-suited for privacy-sensitive applications. The HDIA dataset contains data collected from IR camera images and wearable armband sensors, both of which preserve subject anonymity by not revealing identifiable visual information.
- The HDIA dataset is distinctive in that it provides both visual and non-visual data for HAR. It includes data captured from various viewing angles and across different sequence lengths, enhancing its versatility for HAR applications.
- The HDIA dataset is publicly available on the GitHub <https://github.com/ai4r/KETI-actiondetection/>. We hope that our contributions will advance future research in the field of HAR.

II. RELATED WORKS

A. MULTIMODAL HAR DATASETS

1) UTD-MHAD

The UTD Multimodal Human Action Dataset (UTD-MHAD) encompasses 27 different actions performed by eight subjects (four females and four males) in an indoor setting. The dataset was collected using both a Microsoft Kinect sensor and wearable inertial sensor. To ensure data reliability,

TABLE 1. Comparison of HDIA dataset and other HAR datasets: (RGB) video, (D)epth, (S)keleton, Infrared (IR), Audio (Au), Ambient sensor (Amb), Object sensor (Obj), Accelerometer(Acc), Gyroscope (Gyr), Magnetometer (Mag), Angular velocity (Ang), Eletromyography (EMG), Inertial measurement unit (IMU), (d)ynamic view.

	Dataset	Modalities	# Sub	# Class	# Samples	# Views
IR(Visual)	NTU RGB+D 120 [25]	RGB,D,S,IR	106	120	114,480	155
	PKU-MMD [26]	RGB,D,S,IR	66	51	21,545	3
	InFAR [16]	IR	40	12	600	2
	IITR-IAR [17]	IR	35	21	1,470	d
Multimodal (Non-visual)	OPPORTUNITY [24]	Acc,Gyr,Mag,Obj,Amb	4	16	701,366	-
	UCI Smartphone [27]	Acc,Gyr	30	6	10,299	-
	PAMAP2 [22]	Acc,Gyr,Mag	9	18	2,844,868	-
	USC-HAD [23]	Acc,Gyr	14	12	840s	-
	SHO [28]	Acc,Gyr,Mag	10	7	630,000	-
Multimodal (Visual+ Non-Visual)	UTD-MHAD [29]	RGB,D,S,IMU	8	27	861	1
	Berkely MHAD [30]	RGB,D,S,Acc,Au	12	11	660	4
	CZU-MHAD [31]	D,S,IMU	7	22	880	1
	C-MHAD [32]	RGB,Acc,Ang	24	12	245	1
	HDIA(ours)	IR,IMU,EMG	47	48	4,306	d

each subject performed each action four times, resulting in 861 data sequences after the removal of three corrupted sequences.

The UTD-MHAD comprises four distinct modalities recorded in three channels: RGB videos, depth videos, skeletal joint positions, and inertial sensor signals. The Kinect camera was positioned on a tripod at a distance of three meters from the subject, with the objective of capturing the entire body in the depth image. The wearable inertial sensor was attached to the right wrist or thigh depending on the specific motion being performed.

The dataset covers a wide range of actions, including specific right arm movements such as “right arm swipe to the left,” “right hand wave,” “right arm throw,” “right hand draw circle,” and more. Additionally, it includes full-body movements like “jogging in place,” “walking in place,” “sit to stand,” and others.

2) BERKELEY MHAD

The Berkeley Multimodal Human Action Database (MHAD) is a dataset that consists of 11 distinct actions performed by 12 subjects (seven males and five females) between the ages of 23 and 30. Each subject performed each action five times, resulting in approximately 660 motion sequences and a total recording time of approximately 82 minutes.

The Berkeley MHAD captures dynamic body movements by using multiple sensing modalities. The dataset incorporated an optical motion capture system, four multi-view stereo vision cameras, two Microsoft Kinect cameras, six wireless accelerometers, and four microphones. These sensors provide a rich set of data for analyzing and understanding human actions.

Activities in the Berkeley MHAD dataset primarily involve dynamic movements of the upper and/or lower extremities. Examples of these actions include “jumping in place,” “jack jumping,” “throwing a ball,” and more. However, certain activities such as “hand waving,” “clapping,” and others specifically focus on the dynamic movements of the upper extremities.

3) CZU-MHAD

The Changzhou University Multimodal Human Action Dataset (CZU-MHAD) is a comprehensive dataset consisting of 22 different actions performed by five male subjects. Each action was repeated eight times, resulting in 880 samples.

The CZU-MHAD dataset was captured using a Microsoft Kinect camera and ten wearable sensors. The dataset incorporates a variety of modalities, including depth video sequences, skeletal joint coordinates, three-axis acceleration data, and three-axis angular velocity data. The diversity of these modalities provides a robust foundation for the analysis and comprehension of human actions.

The 22 actions included in the CZU-MHAD dataset were representative of a diverse range of gestures and activities. Examples of the actions included in the CZU-MHAD set are common gestures such as “draw x” and “draw circle,” daily activities such as “sur place,” “clap,” and “bend down,” as well as training actions such as “left body turning movement” and “left lateral movement.” The dataset provides a comprehensive collection of actions that can be utilized for the study of human actions and development of robust recognition models.

B. KEY BENEFITS OF A PROPOSED DATASET

This study presents a novel multimodal HAR dataset that emphasizes participant privacy. In contrast to other datasets in HAR that rely primarily on RGB video data for visual data, we utilized IR video data as the primary visual modality to ensure the privacy of participants. Moreover, we integrated IMU and EMG data from wearable sensors, thereby enabling the capture of both global information derived from video and local information derived from sensor data. This multimodal approach permits a more comprehensive understanding of human activity, resulting in a diverse and robust dataset.

The dataset was collected in a real-life apartment setting. This allowed participants to act more naturally and realistically, with minimal behavioral constraints, thereby ensuring that the data were more reflective of everyday activities and, thus, were more applicable to real-world scenarios.



FIGURE 1. Sensors (a) FLIR Boson 320A50 thermal imaging camera, (b) MYO armband sensor.

Furthermore, to enhance the adaptability of HAR, we incorporated dynamic camera views to document human activities from diverse angles, thereby increasing the reliability of the recognition across a range of viewpoints. Compared to existing multimodal datasets, which typically encompass up to 27 classes, our dataset notably expanded the number of behavioral classes to 48, collected from 47 participants.

III. HDIA DATASET

A. DATA COLLECTION

Both a thermal camera and armband sensor was employed to gather multimodal data. The IR camera captures the action image information while minimizing the potential for face identification. Concurrently, the armband sensor serves as a non-visual wearable sensor that captures local arm movements. This approach enables the collection of data that provides both global and local movement patterns, which are beneficial in situations where visual perception is limited and actions may appear similar.

Figure 1 illustrates the IR camera and armband sensor used for data collection. To capture IR data, we employed the FLIR Boson 320A50 thermal imaging camera, which features a thermal sensor capable of recording images at a resolution of 640×512 pixels with a frame rate of 8 fps.

For non-visual data, we used the MYO armband developed by Thalmic Labs. This device provides two distinct data streams: a 9-axis IMU dataset and an 8-channel EMG dataset. Participants were instructed to wear the armband such that the embedded IMU sensor was aligned with their thumb for consistent orientation. The EMG sensors captured muscle activation signals at a sampling rate of 200 Hz, while the IMU recorded forearm motion data at 50 Hz. All data were transmitted via a Bluetooth Low Energy (BLE) module [35] and saved in Comma-Separated Values (.csv) format with synchronized timestamps.

We rented a real apartment for data collection to ensure a natural and realistic setting. As shown in Figure 2, participants performed everyday activities in various areas of the apartment, including the living room, kitchen, bathroom, and bedroom. Cameras were positioned at a height of 70–120 cm to approximate a robot's viewpoint and to accommodate movement within the furnished space. Participants wore armband sensors on their wrists throughout the sessions.

Recording in a real-life environment allowed participants to engage in their daily routines with minimal constraints,

fostering more authentic behavioral data. To guide the data collection process, participants were provided with written instructions outlining the target actions, but they were encouraged to carry out the tasks in their own manner. Additionally, data were recorded from multiple viewpoints—including front, side, and rear angles. The videos are captured from rear views such as *washing dishes* or *washing their face*, rather than frontal views, which provide insight into human actions.

B. DATA ORGANIZATION

The HDIA dataset comprises 48 distinct actions performed by 47 participants, including 22 males and 25 females. Each subject performed each action approximately three times, providing a diverse and representative dataset for HAR.

To define the action classes, we conducted field observations of older adults in their home environments. Specifically, we visited the homes of 53 individuals aged 70 and above and carefully documented their daily routines from morning to night. Based on these observations, we identified 48 commonly performed activities encompassing a broad range of behaviors, including eating, grooming, household chores, and social interactions. To ensure that the dataset captured realistic behavioral variability, we recruited 47 participants to perform the identified actions. The complete list of 48 actions is provided in Table 2. The samples of IR image are shown in Figure 3. As a result, the HDIA dataset offers valuable insights into the behavioral patterns of older adults and supports a deeper understanding of their everyday lives.

The data processing pipeline involved recording, annotating, and segmenting the raw inputs, as illustrated in Figure 4. Thermal camera data and armband signals were streamed and saved with synchronized timestamps. During the annotation phase, video recordings were reviewed and labeled with metadata, including start time, end time, posture, action, and the hand used. All data files were named following a predefined naming convention, and the corresponding armband signals were segmented based on the annotated time intervals.

Our dataset includes both IR video data and armband sensor data, as illustrated in Figure 5. Careful preprocessing was conducted to exclude incomplete or corrupted samples, such as instances with missing modalities or where the participant's full body was not properly captured. The final dataset comprises 4,306 data sequences, with a total recording duration of approximately 11 hours and 30 minutes. On average, each action class contains 89.71 samples, with a standard deviation of 29.67. The number of instances per action ranges from 28 to 163. Sequence durations vary significantly, ranging from 2 to 37 seconds, with an average duration of 9.61 seconds and a standard deviation of 3.93 seconds. The shortest action is putting on/taking off glasses (class 20), while the longest is reading a newspaper (class 31). The distribution of action classes and sequence durations is visualized in Figure 6.

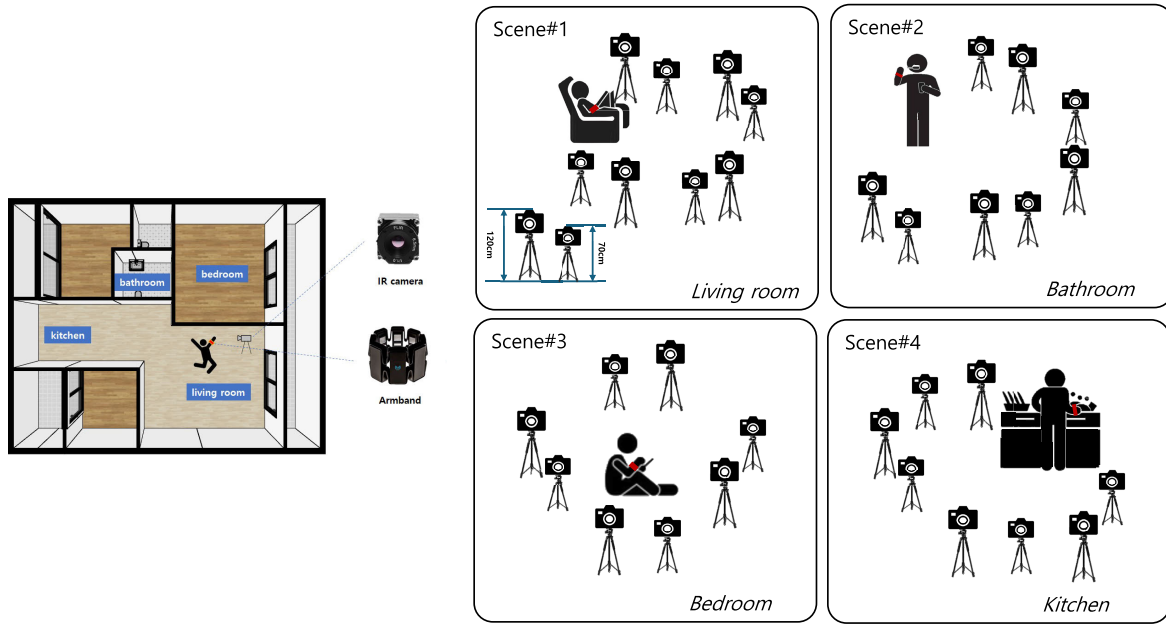


FIGURE 2. Layout of the data collection setup. The experimental environment consists of four distinct areas: the living room, kitchen, bathroom, and bedroom. IR cameras were installed at two different heights to capture human activities from multiple perspectives. A wearable armband—highlighted in red—was used to collect motion and activity data from each participant. Scenes (#1–#4) depict various activity scenarios corresponding to each area of the environment.

TABLE 2. List of 48 actions included in the HDIA dataset. Each action is assigned a unique ID and represents a diverse range of everyday activities, including eating, grooming, household chores, and social interactions.

ID	Action	ID	Action	ID	Action
1	eating food with a fork	17	drying hair	33	talking on the phone
2	pouring water into a cup	18	wearing jacket	34	playing with a cell phone
3	taking medicine	19	taking off a jacket	35	using a computer
4	drinking water	20	wearing on glasses/taking off glasses	36	smoking
5	putting/taking food in/from the fridge	21	washing the dishes	37	clapping
6	trimming vegetables	22	vacuuming the floor	38	wiping face
7	peeling fruit	23	mopping the floor with kneeling down	39	doing bare-handed gymnastics
8	using a gas stove	24	wiping off the dining table	40	shoulder self-massage
9	cutting vegetable on the cutting board	25	rubbing up furniture	41	handshaking
10	brushing teeth	26	folding/unfolding a duvet	42	hugging other person
11	washing hands	27	hand-wash clothes	43	waving a hand
12	washing face	28	hanging out laundry	44	beckoning to come
13	wiping face/hair with a towel	29	using a remote control	45	pointing with a finger
14	putting on cosmetics	30	reading a book	46	opening doors and walking through
15	putting on lipstick	31	reading a newspaper	47	getting up and standing up
16	brushing hair	32	handwriting	48	lying down

We organized and labeled the data using a standardized action sequence naming protocol in the format “Axxx_Pxxx_Gxxx,” where A denotes the action class, P the participant ID, and G the trial number, accounting for variations in posture and background [36]. Detailed information on the annotation and segmentation procedures for each modality is provided in the accompanying documentation. The HDIA dataset, along with the annotation manual, is publicly available on GitHub at <https://github.com/ai4r/KETI-actiondetection/>.

C. DATA PROCESSING

The IR video data is represented as a sequence of frames $I = f_1, f_2, \dots, f_i$ which serve as input to a video-based action

recognition network. This network extracts the IR feature vectors $F_{IR} \in \mathbb{R}^{T \times N_{dim}}$, where T represents the temporal dimension and N_{dim} denotes the feature dimension.

In the case of wearable sensor data, which includes both IMU and EMG signals, a sliding window is applied to partition the raw data into T input sequences, denoted as $S = \{w_1, w_2, \dots, w_T\}$. To improve the quality of the raw EMG signals, we apply additional signal processing steps to smooth extreme fluctuations around zero and reduce noise. Specifically, the EMG signals are amplified, rectified, and filtered to extract their envelope. This is achieved using a second-order low-pass filter (LPF) with a cutoff frequency of 2 Hz. A comparison between the raw and processed EMG signals is presented in Figure 7.

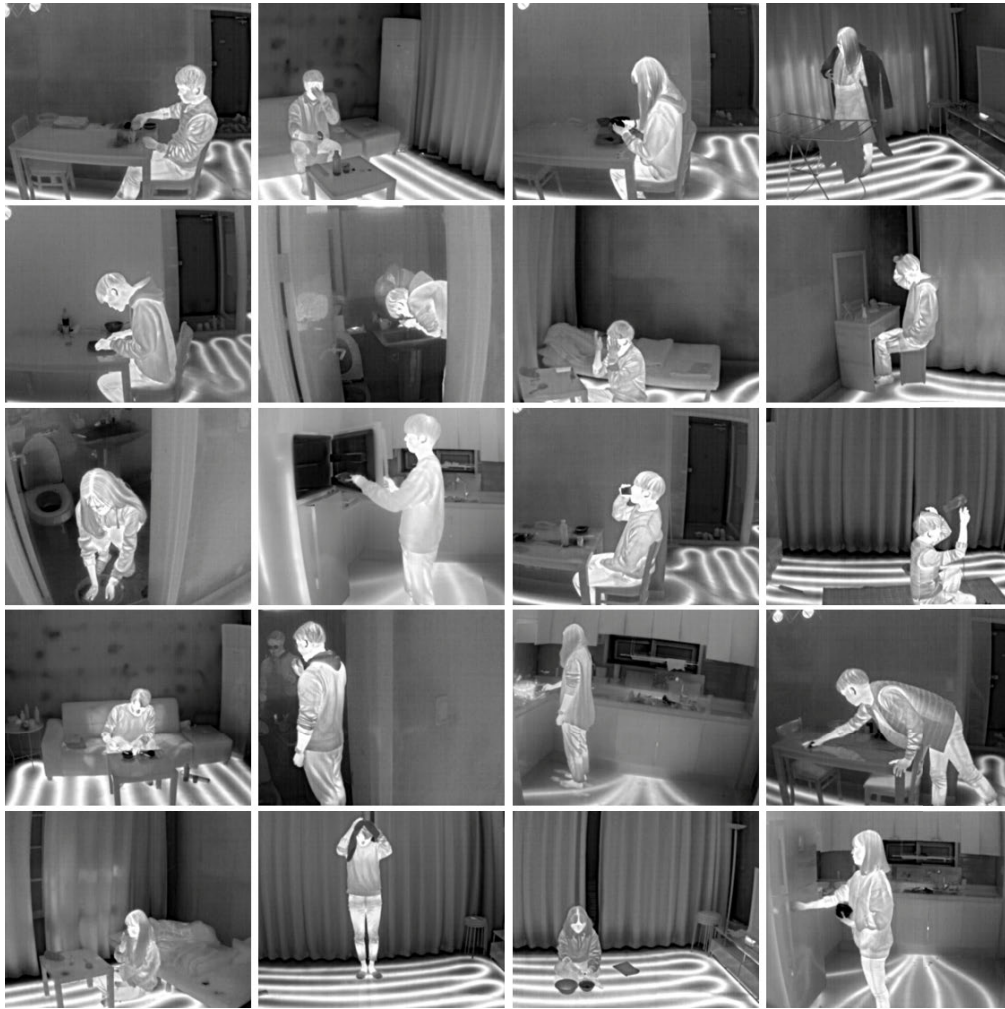


FIGURE 3. IR image frames in HDIA dataset. 47 participants performed 48 types of daily indoor actions from various views.

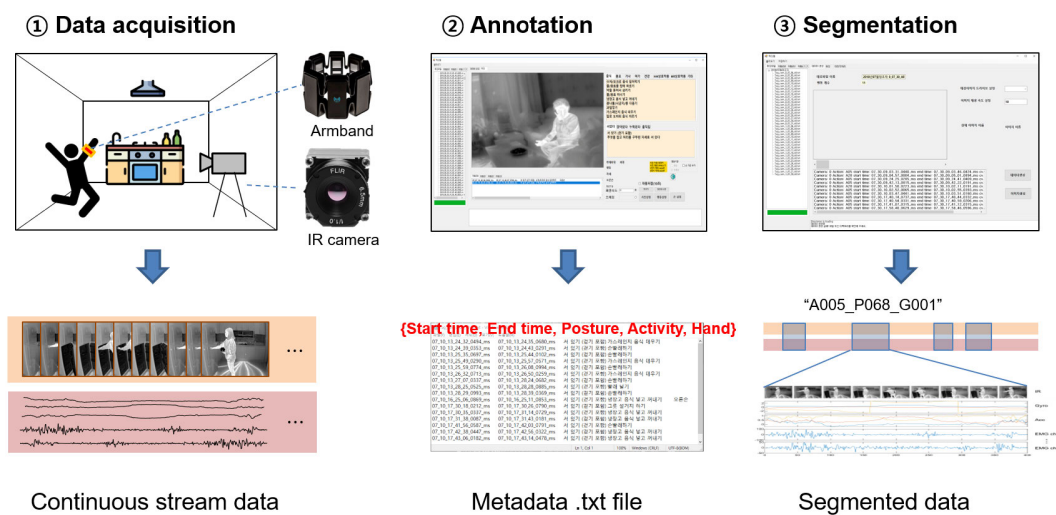


FIGURE 4. Workflow for dataset collection, annotation, and segmentation. Data is acquired via an IR camera and armband, annotated with metadata, and segmented into labeled sequences with synchronized video and sensor data for analysis.

Each sequence, denoted as s_i , is then processed using a Long Short-Term Memory (LSTM) network [18] to extract

features. The resulting feature representations are expressed as $F_S \in \mathbb{R}^{T \times N_{\text{dim}}}$, where T represents the sequence

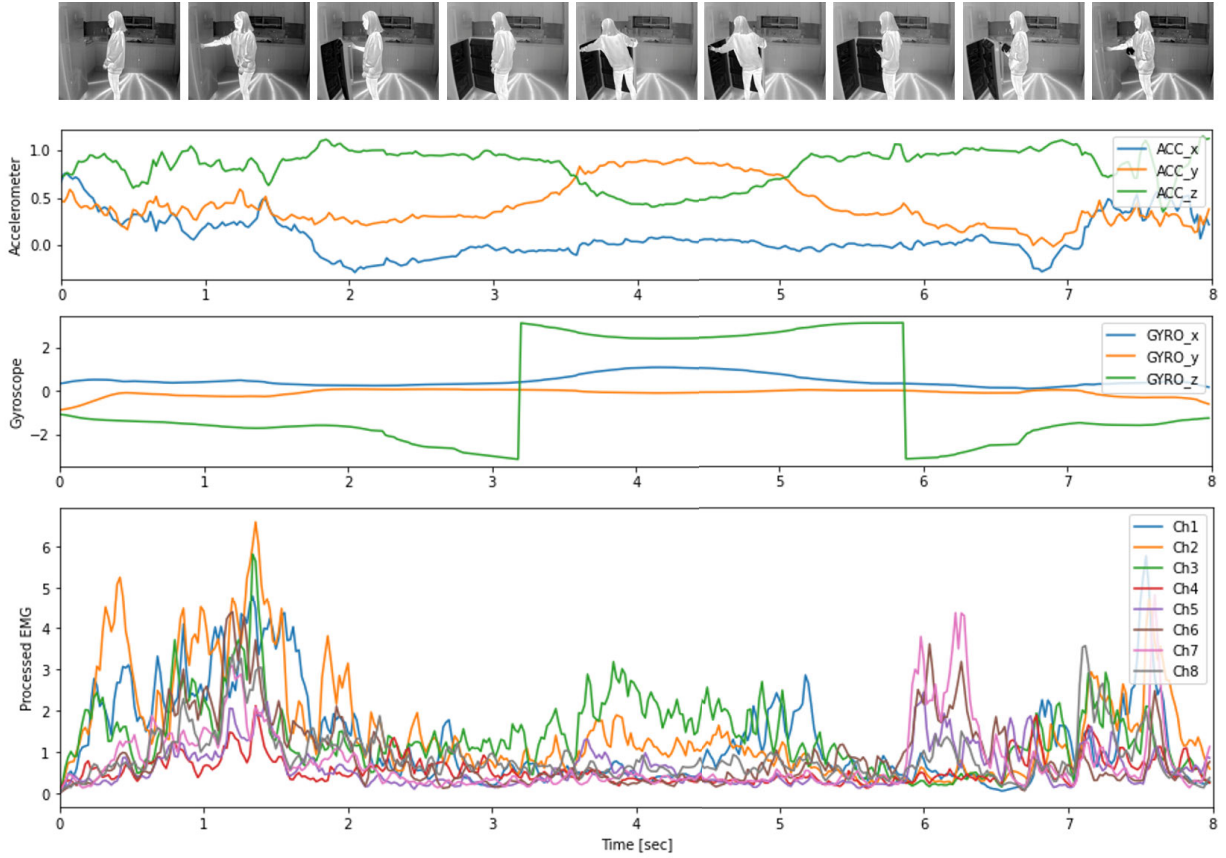


FIGURE 5. Example of time-synchronized final processed data from the HDIA dataset. The figure presents synchronized data streams, including IR images, raw IMU sensor data (ACC_x , ACC_y , ACC_z , $GYRO_x$, $GYRO_y$, $GYRO_z$), and processed EMG signals from 8 channels.

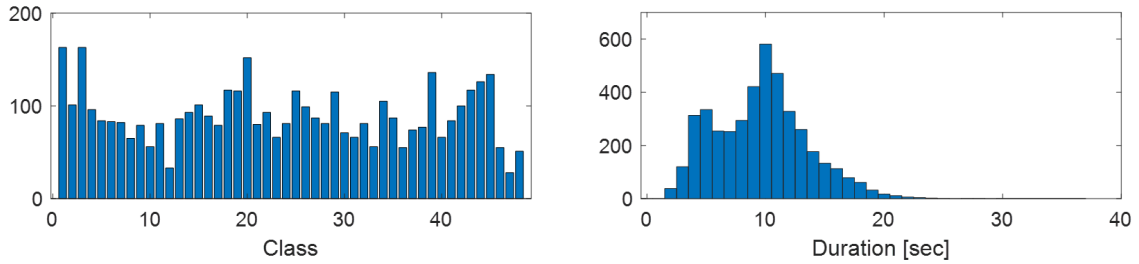


FIGURE 6. Statistics of HDIA dataset. Each class comprises an average of 89.71 instances (left), with an average duration of 9.61 seconds (right).

length and N_{dim} denotes the dimensionality of the extracted features.

IV. BENCHMARK ON HDIA DATASET

In this study, we introduce a baseline HAR method for performance comparison. We design a multimodal fusion network based on the Transformer architecture [19]. In addition, we present a HAR benchmark built on the HDIA dataset to facilitate standardized evaluation and comparison of multimodal HAR approaches.

A. BASELINE METHOD

The feature vectors F_{IR} and F_S are concatenated to form the input to the transformer $Z_{in} \in \mathbb{R}^{T \times 2N_{dim}}$. The transformer architecture serves to fuse spatial-temporal information

extracted from both IR and sensor data. By using self-attention mechanisms, the transformer captures intricate relationships and dependencies between elements in both modalities.

The two layers of self-attention operation are applied to Z_{in} to refine the representations. This process helps capture the complex dependencies between the IR and armband sensor features, as well as across time. The self-attention mechanism is formulated as follows

$$Z_{in} = \text{Concat}(F_{IR}, F_S) \quad (1)$$

$$Q_h = Z_{in} W_{Q,h}, \quad K_h = Z_{in} W_{K,h}, \quad V_h = Z_{in} W_{V,h}, \quad (2)$$

where $W_{Q,h} \in \mathbb{R}^{2N_{dim} \times h_{dim}}$, $W_{K,h} \in \mathbb{R}^{2N_{dim} \times h_{dim}}$, and $W_{V,h} \in \mathbb{R}^{2N_{dim} \times h_{dim}}$ are projection matrices, where $h_{dim} = \frac{2N_{dim}}{H}$. The

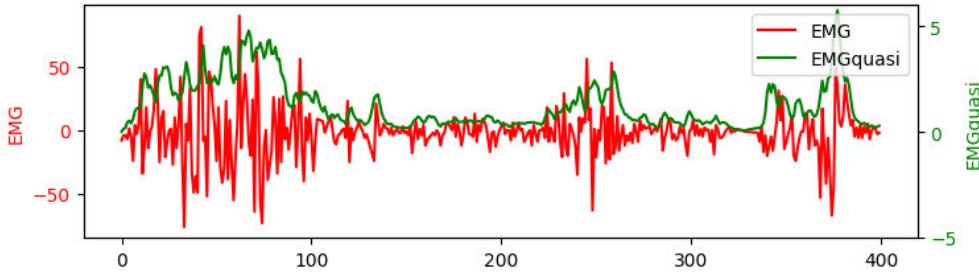


FIGURE 7. Comparison of raw EMG signals (red) and processed EMG signals (EMGquasi, green). Raw EMG is noisy and variable, while EMGquasi smooths it to highlight key muscle activity patterns.

index h refers to the current attention head, and H is the total number of heads. The multi-head self-attention operation is computed as

$$\text{head}_h = \text{softmax}\left(\frac{QhK_h^T}{\sqrt{d_k}}\right)V_h \quad (3)$$

$$Z_{out} = [\text{head}_1; \text{head}_2; \dots; \text{head}_H]W_O, \quad (4)$$

where d_k is a scaling factor for normalization, and $W_O \in \mathbb{R}^{2N_{dim} \times 2N_{dim}}$ is a learnable weight matrix. The resulting output, Z_{out} , has dimensions $T \times 2N_{dim}$.

After the multi-head self-attention step, the output Z_{out} is added element by element to another representation, meaning values at the same positions are summed. This operation is intended to enhance cross-modal interactions, allowing more effective integration of information from the IR and sensor data. Subsequently, the merged feature representation is conveyed through a multilayer perceptron (MLP) for classification.

B. HDIA BENCHMARK

In this study, we adopted a cross-subject protocol — splitting the dataset by subject identity — to ensure a robust evaluation of the proposed methods under realistic conditions. Specifically, 37 subjects were used for training and a distinct set of 10 subjects was reserved for testing, eliminating any overlap between training and test groups. This setup yields a more accurate measure of model generalizability. For unimodal experiments, we applied the TSM [9] to IR data, while the DeepConvLSTM model [37] was trained on armband sensor data. These two approaches serve as baselines, each using a single sensing modality.

We then considered a multimodal fusion strategy to evaluate the benefits of combining IR and armband sensor data. First, we tested the Weighted Sum Method (WSM) [38]. Then, we presented a proposed multimodal fusion network that jointly learns and aggregates features from both modalities. For IR data, ResNet-50 [39] and MobileNet-V2 [40] were compared in both standard and optimized configurations, ensuring that the observed performance gains are not tied to a single network architecture. Overall, this benchmark design evaluates dataset performance through a benchmark

model using cross-subject data splitting, uni-mode approach, fusion methods, and various backbone architectures.

V. EXPERIMENTS

The performance of our HDIA dataset was evaluated for a range of networks to confirm its suitability for use. To evaluate the utility of the dataset, it was benchmarked against well-established methods, and the correlation performance of the data was quantified using the proposed network. A performance evaluation was conducted using Top-1 Accuracy and F1-Score, as these metrics provide complementary insights into the model's performance. The Top-1 accuracy quantifies the capacity of the model to accurately categorize the initial prediction, which is critical for tasks. In contrast, the F1-score provides a more comprehensive evaluation of performance, particularly in datasets with class imbalances or variable label distributions, by balancing the precision and recall.

A. SETUP

An intensive empirical validation was conducted to identify the optimal set of hyperparameters, specifically the learning rate, batch size, and number of epochs—for maximizing performance. Training of the fusion network was conducted using the stochastic gradient descent (SGD) optimization technique, with a learning rate of $5e^{-3}$ and a momentum of 0.9. The training process was executed for 80 iterations, and the learning rate was adjusted in real-time based on the validation performance to maintain the stability in the fusion process. Batch regularization and dropout layers were employed to prevent overfitting, and weight decay was applied to regularize the model.

To perform feature extraction from IR data, the ResNet50 model [39] was employed as the backbone network. The deep residual network architecture, which was pre-trained on ImageNet, was fine-tuned on the HDIA dataset with a particular focus on the extraction of spatial and temporal features from IR video frames. The training of the IR feature extractor required approximately 15 hours. In contrast, armband data, which consist of low-dimensional sensor readings, were processed using a lightweight feature extractor. The extractor was trained for approximately 2 hours, owing to the sequential structure of the armband sensor data.

TABLE 3. Comparison of unimodal and multimodal methods on the HDIA dataset using Top-1 Accuracy and F1-Score, highlighting performance improvements through multimodal fusion.

Method		Modality	Top-1 Accuracy (%)	F1-Score	Min. F1-Score (no. class)
TSM [9]	ResNet-50	IR	86.414	0.856	0.53 (33)
	MobileNet-V2		84.185	0.835	0.58 (33)
DeepConvLSTM [37]		Inertial	66.674	0.661	0.35 (6)
WSM [38]	ResNet-50	IR + Inertial	87.459	0.870	0.57 (33)
	MobileNet-V2		84.928	0.841	0.59 (36)
Ours	ResNet-50	IR + Inertial	90.101	0.901	0.51 (32)
	MobileNet-V2		89.835	0.899	0.50 (32)

The complete training and testing process was executed on a single NVIDIA GeForce GTX 2080Ti GPU, with a batch size of 8 to achieve an optimal balance between computational efficiency and convergence stability. The loss function utilized was the cross-entropy loss.

B. PERFORMANCE EVALUATION

Table 3 presents a detailed performance evaluation of the unimodal and multimodal fusion approaches for the HDIA dataset. As illustrated in Table 3, all methods, achieved a Top-1 Accuracy of over 80%, with the partial exception of the armband-only model, achieved a Top-1 Accuracy of over 80%, indicating that the HDIA dataset is both stable and sufficient for benchmarking the proposed multimodal approach. In the multimodal fusion scenario, both the WSM and proposed fusion network showed improved performance compared to the unimodal model. However, the improvement observed for the WSM was relatively minor, indicating that a simplified fusion technique may not be sufficient to fully leverage the complementary information from both modalities.

In contrast, the proposed multimodal fusion network exhibited a notable enhancement in performance, with a 4% gain over unimodal IR data, further supporting the validity of our architectural approach for combining IR and armband data. This indicates that the fusion of IR and armband sensor data significantly amplifies the dimensionality of the feature space, thereby achieving more optimal task-specific performance than would be possible with either modality in the absence of the other. It is crucial to note that the two modalities provided by the HDIA dataset contain complementary information, and that the performance is observed to improve when the armband sensor is fused with the IR data.

Moreover, the F1-Score exhibited consistent values above 0.6 for all the methods, thereby further confirming the stability of the models. At a minimum, all models except the armband-only approach achieved F1-scores above 0.5, thereby providing further valuable insight into the robustness of each approach. This reinforces the notion that while armband data alone may not provide sufficient discriminatory

TABLE 4. Performance comparison on benchmark dataset using baseline method.

Dataset	Modalities	Subjects	Activities	Top-1 Accuracy (%)	F1-Score
Berkeley-MHAD	RGB,D.S,Acc,Au	12	11	100	1.0
UTD-MHAD	RGB,D.S,IMU	8	27	96.262	0.96
HDIA Dataset	IR,IMU,EMG	47	48	90.101	0.90

power for this task, it nevertheless has value in a multimodal environment.

We evaluated the efficacy of the baseline method by employing it with an additional multimodal benchmark dataset. The findings presented in Table 4, demonstrate that the baseline method consistently achieved high accuracy, with 100% Top-1 accuracy on the Berkeley-MHAD dataset and 96.26% Top-1 accuracy on the UTD-MHAD dataset. These results underscore the model's adaptability, indicating that its performance is not exclusively optimized for HDIA. The substantial accuracy across diverse datasets underscores the robustness of the baseline method in multimodal HAR, demonstrating its capacity to effectively adapt to varied sensor inputs and data distributions.

Figure 8 shows the classification accuracy for each activity evaluated using the baseline method. The accuracy levels ranged from 42%, representing the lowest level of accuracy, to 100%, representing the highest level of accuracy. Most activities demonstrated an accuracy level that exceeded 90%. It is noteworthy that specific actions such as "lie down," "get up and stand up," "hugging another person," and "handshaking," demonstrate near-perfect accuracy, suggesting that the model exhibits a high degree of proficiency in recognizing these movements. By contrast, activities involving more subtle hand movements, such as "playing with a cell phone" and "typing on a keyboard," demonstrate relatively lower accuracy, suggesting that these tasks may present more challenges for classification. The results indicate that the baseline method generally exhibits robust performance across a broad range of activities and displays consistently high accuracy in most cases. Variation in performance across different activities underscores the potential for enhancement, particularly in the context of fine-motor tasks that demand detailed hand interactions.

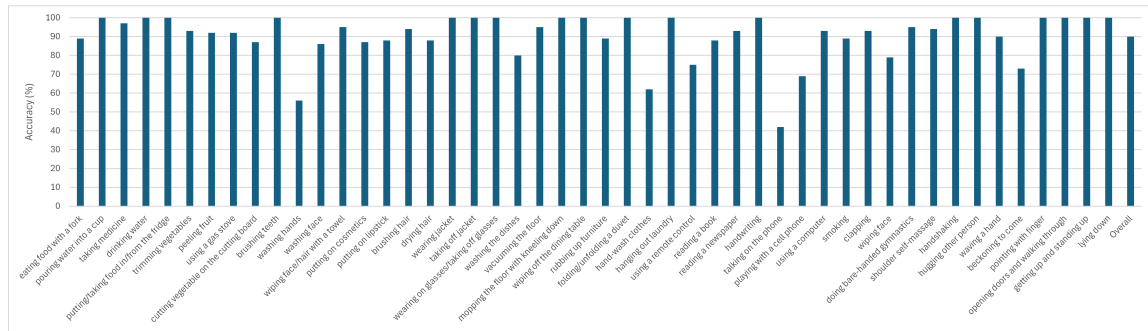


FIGURE 8. Class-specific and overall accuracy of 48 HDIA dataset human actions using the baseline method.

This evaluation further supports the effectiveness of multimodal approaches in HAR. While each modality contributes unique strengths, their combination within a well-designed fusion network enhances overall performance. Moreover, the HDIA dataset, with its diverse range of data streams, has proven to be a valuable benchmark for assessing and validating multimodal learning techniques, demonstrating its applicability for advancing research in this field.

VI. CONCLUSION

This study introduced the HDIA dataset, a comprehensive multimodal resource designed to enhance HAR research, with a particular emphasis on elderly individuals' daily behaviors. The dataset consists of 48 common indoor activities recorded from 47 subjects using IR cameras and wearable armband sensors. The data collection process was carefully designed to ensure privacy protection by utilizing non-invasive sensors that do not reveal personal identity. Additionally, by capturing activities in real-world residential settings, the dataset provides a more natural and diverse representation of behaviors than conventional laboratory-based datasets.

To demonstrate its practical applicability, we implemented a benchmark fusion model integrating the IR and IMU sensor data, achieving an accuracy of approximately 90%. While this result highlights the effectiveness of multimodal data fusion, further research is needed to systematically compare HDIA with existing HAR datasets, particularly in terms of environmental diversity and elderly specific activity patterns. Future studies could focus on conducting comprehensive qualitative and quantitative evaluations, as well as exploring new methods for improving activity recognition performance in elderly care applications.

The HDIA dataset has significant potential for advancing applications in robotics, smart home automation, and assistive technologies, as it provides privacy-conscious, multimodal sensor data that can support the development of adaptive and context-aware HAR models tailored to the needs of elderly individuals.

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